

Exercise 2 SQL Aggregate functions & OP

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* The Students table

~~1. List~~

Students

student_id	name	age	department
1	Alice	20	IT
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

1. List all distinct departments in the Students table.

```
SELECT DISTINCT department
FROM students;
```

department
IT
HR
Finance

2. Get the average age of Students per department

```
SELECT department
      AVG (age) AS avg_age
```

```
FROM students
```

```
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22.0
Finance	23.0

3. Show departments with more than 1 student

```
SELECT department
```

```
      COUNT (student_id) AS student_count
```

```
FROM students
```

```
GROUP BY department
```

HAVING COUNT(student-id) > 1;

department	student-count
IT	2
HR	2

4. Get all Students whose age is between 21 and 23

~~SELECT department~~
~~COUNT (student-id) AS student-count~~ ~~FROM students~~
~~FROM students~~ ~~WHERE age BETWEEN 21 AND 23;~~
 GROUP BY

student-id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	Finance
5	Eve	22	HR

5. List all Students in the IT or HR department who are older than 21.

SELECT student-id, name, age, department
 FROM students
 WHERE (department = 'IT' OR department = 'HR') AND age > 21

student-id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

* The courses table

courses

course-id	course-name	department	credits
101	SQL Basics	IT	3
102	Python	IT	4
103	Data Science	IT	4
104	Excel	Finance	2
105	Statistics	HR	3

6. Show total credits per department, only for departments with more than 5 total credits.

```
SELECT department
Sum (credits) AS total_credits
FROM courses
GROUP BY department
HAVING
```

department	total_credits
IT	11

7. List all courses that do not have 4 credits

```
SELECT course_id, course_name, department, credits
FROM courses
WHERE credits <> 4;
```

course_id	course_name	department	credits
101	SQL Basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

8. Show the top 3 courses by credits in descending order.

```
SELECT course_id, course_name, credits
FROM courses
ORDER BY credits DESC
LIMIT 3;
```

course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL Basics	3

* The enrollment table

8. enrollments

enrollment_id	student_id	course_id	grade
1	1	101	85
2	2	102	78
3	3	103	90
4	4	104	88
5	5	105	82

9. Get the maximum, minimum, and average grade across all enrollments

SELECT MAX(grade) AS max_grade,
 AVG(grade) AS avg_grade

FROM enrollments;

max_grade	min_grade	avg_grade
90	78	84.6

10. Count how many enrollments exist per course.

SELECT course_id,

COUNT(enrollment_id) AS enrollment_count

FROM enrollments

GROUP BY ~~department~~; course_id;

department	course_id	enrollment_count
IT	101	1
	102	1
	103	1
	104	1
	105	1

11. Find total salary and total bonus per department

SELECT department,

SUM(salary) AS total_salary

SUM(bonus) AS total_bonus

FROM salaries

GROUP BY ~~salary~~ department;

department	total-salary	total-bonus
IT	122 000	105 00
HR	109 000	7500
Finance	70 000	6000

* The salaries table

salaries

employee-id	name	department	salary	bonus
1	Tom	IT	60 000	5000
2	Jerry	HR	55 000	4000
3	Spike	Finance	70 000	6000
4	Tyke	IT	62 000	5500
5	Butch	HR	54 000	3500

12 Show departments where average salary is 55 000

SELECT department,
AVG(salary) AS avg-salary

FROM salaries

GROUP BY department

HAVING AVG(salary) > 55 000;

department	avg-salary
IT	61 000
Finance	70 000

13. List employees whose salary plus bonus is greater than 65 000.

SELECT employee-id, name, salary, bonus, (salary + bonus) AS total-compensation
FROM salaries
WHERE (salary + bonus) > 60 000;

employee-id	name	salary	bonus	total-compensation
1	Tom	60 000	5000	65 000
3	Spike	70 000	6000	76 000
4	Tyke	62 000	5500	67 500

* The projects table

15. Show total and average budget per department. Only include departments with average budget

~~SELECT project-id, project-name, department, budget~~

~~SELECT department~~

~~(SUM(budget) AS total-budget~~

~~AUG(budget) AS avg-budget~~

~~FROM projects~~

~~GROUP BY department~~

~~HAVING AVG(budget) > 70 000;~~

department	total-budget	avg-budget
IT	270 000	135 000
Finance	80 000	80 000

15. List all projects with budgets between 50 000 and 120 000, excluding the Marketing department.

SELECT project-id, project-name, department, budget

FROM projects

WHERE (budget BETWEEN 50000 AND 120000) AND department <> 'Marketing'

project-id	project-name	department	budget
1	AI APP	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	50 000

END